



Post-Quantum Cryptography Risk Assessment

Quantum Vulnerability Assessment Report

Generated: June 07, 2026 13:40 UTC

Executive Summary

50

MEDIUM RISK

Moderate quantum vulnerability risk. Some systems require attention before Q-Day.

1

Total Endpoints

0

Vulnerable Endpoints

0.0%

Vulnerability Rate

Assessment Overview:

Assessment of 1 endpoint(s) reveals a moderate quantum vulnerability risk with an overall score of 50/100. All scanned endpoints appear to use quantum-resistant configurations. Continue monitoring for new vulnerabilities and emerging standards.

Key Findings

No critical or high priority issues were identified in this assessment.

Why This Matters

What is "Harvest Now, Decrypt Later" (HNDL)?

HNDL is a threat model where adversaries collect encrypted data today with the intention of decrypting it once quantum computers become capable of breaking current encryption algorithms.

Why This Matters:

- Data with long-term sensitivity (medical records, financial data, state secrets) is at risk even before quantum computers exist.
- RSA and ECC-based key exchange, used in most TLS connections today, will be broken by Shor's algorithm running on sufficiently powerful quantum computers.
- Organizations should begin transitioning to post-quantum cryptography (PQC) now to protect data being transmitted today.

Timeline: While estimates vary, many experts predict cryptographically relevant quantum computers (CRQCs) could emerge within 10-15 years. Given the time required for cryptographic migrations, organizations should start planning now.

Recommendations

Recommended Actions

- Develop a comprehensive cryptographic inventory
- Create a post-quantum migration roadmap
- Monitor NIST post-quantum cryptography standards
- Implement crypto-agility in system design

This report was generated by ShaheenShield - Post-Quantum Cryptography Risk Assessment Platform

Report ID: ACDF3B1E | Scan ID: 2 | Generated: June 07, 2026 13:40 UTC

© 2026 ShaheenShield. All rights reserved.

TLS Scan Findings

The following table presents the detailed TLS configuration for each scanned endpoint:

Endpoint	TLS Version	Key Exchange	Cipher	Risk Score	Status
shaheenshield.com:443	N/A	N/A	N/A	50/100	Unknown

Risk Score Breakdown

The overall risk score is calculated based on the following component scores:

Component	Average Score	Weight	Impact
Key Exchange	0/100	35%	Low Impact
TLS Version	0/100	20%	Low Impact
Cipher Suite	0/100	15%	Low Impact
Hash Algorithm	0/100	15%	Low Impact
Signature Algorithm	0/100	15%	Low Impact

Configuration Distribution

TLS Version Distribution

TLS Version	Count	Percentage
Unknown	1	100.0%

Key Exchange Distribution

Key Exchange	Count	Quantum Status
Unknown	1	Unknown

Detailed Risk Assessment

This section provides a detailed breakdown of the quantum vulnerability assessment for each endpoint.

shaheenshield.com:443

MEDIUM (50/100)

Component	Value	Status	Score
-----------	-------	--------	-------

Detailed Recommendations

Migration Priority: Based on your assessment score of 50/100, we recommend a moderate priority approach to post-quantum migration.

Immediate Actions (0-6 months)

- **Cryptographic Inventory:** Create a complete inventory of all systems using public-key cryptography, especially those handling sensitive or long-lived data.
- **Risk Assessment:** Prioritize systems based on data sensitivity and exposure to HNDL attacks.
- **Vendor Engagement:** Contact vendors of critical systems about their post-quantum roadmaps.

Short-term Actions (6-18 months)

- **Crypto-Agility:** Implement cryptographic agility in system designs to facilitate future algorithm transitions.
- **Hybrid Solutions:** Consider deploying hybrid classical/post-quantum solutions for highest-risk systems.
- **Testing:** Begin testing NIST-approved post-quantum algorithms (ML-KEM, ML-DSA, SLH-DSA) in non-production environments.
- **TLS 1.3:** Ensure all systems support TLS 1.3, which will be the foundation for post-quantum TLS.

Long-term Actions (18+ months)

- **Migration Execution:** Execute phased migration to post-quantum algorithms based on risk prioritization.
- **Continuous Monitoring:** Implement ongoing monitoring for cryptographic configurations across all systems.

- **Standards Tracking:** Stay current with evolving NIST post-quantum standards and industry best practices.
- **Training:** Ensure development and security teams are trained on post-quantum cryptography concepts and implementation.

Assessment Methodology

This assessment evaluates the quantum vulnerability of TLS/SSL configurations by analyzing:

Component	What We Check	Quantum Impact
Key Exchange	RSA, ECDHE, DHE, etc.	RSA and ECC-based methods are vulnerable to Shor's algorithm
Signature Algorithm	RSA, ECDSA, Ed25519, etc.	Traditional signatures can be forged with quantum computers
Symmetric Cipher	AES-128, AES-256, ChaCha20	AES-256 provides adequate quantum resistance
Hash Algorithm	SHA-256, SHA-384, SHA-1	SHA-256+ provides sufficient quantum resistance
TLS Version	TLS 1.3, 1.2, 1.1, 1.0	Older versions have weaker security properties

Risk Scoring

Risk scores are calculated based on:

- **Component Classification:** Each cryptographic component is classified as Quantum-Broken, Quantum-Vulnerable, or Quantum-Resistant
- **Weighted Scoring:** Key exchange has highest weight (most critical for HNDL)
- **Aggregation:** Overall score considers worst-case and average scenarios

Score Range	Risk Level	Interpretation
0-25	LOW	Minimal quantum vulnerability
26-50	MEDIUM	Moderate risk, planning recommended
51-75	HIGH	Significant risk, action needed
76-100	CRITICAL	Urgent action required

Disclaimer

This assessment provides a point-in-time evaluation of TLS configurations for quantum vulnerability. The assessment is based on publicly observable cryptographic parameters and does not include:

- Internal network configurations
- Application-layer cryptography
- Data-at-rest encryption
- Hardware security modules (HSMs)
- Other cryptographic implementations

Organizations should conduct comprehensive cryptographic audits to fully understand their quantum vulnerability posture.

This report was generated by ShaheenShield - Post-Quantum Cryptography Risk Assessment Platform

Report ID: ACDF3B1E | Scan ID: 2 | Generated: June 07, 2026 13:40 UTC

© 2026 ShaheenShield. All rights reserved.